COMP 4702 Assignment

Using Machine Learning to Predict Energy in Food

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1.0 Introduction

The dataset, Australia Food Composition Database (AFDC), Release 3, published by Food Standards Australia New Zealand (FSANZ) is the official reference dataset containing detailed information on the nutrient content of foods commonly available in Australia. AFDC is the cumulation of food data collected since the 1980s through multiple different methods. These methods include analytical programs commissioned by FSANZ, analytical projects undertaken by academic or industry groups in collaboration with FSANZ, data provided to FSANZ by academic, industry or other research groups, recipe calculations, overseas food composition tables, the food industry, food label information and imputing from similar foods. (Food Standards Australia New Zealand, 2025)

In this report, we will compare various machine learning techniques such as random forest, bagging and linear models, fine tuning them with accuracy metrics in order to find the best predictor for energy in food. This will help us answer the question:

Can machine learning models outperform the traditional rule-based approach in predicting food energy content from nutrient composition?

2.0 Data Preparation

2.1 Understanding the Dataset

After a quick look, we can see that this dataset contains 1 cover page that outlines the schema, the main table that contains nutrient profiles for all solids and liquids and a subset of the main table that contains nutrient profiles for liquids only.

The main table has 1588 rows and 272 columns. Rows in this table represent the foods that span the Australian diet. This includes fresh produce, meats and seafood, dairy, takeaway items, alcoholic and non-alcoholic drinks, condiments, spices and foods in multiple states (raw/ boiled/ fried/ canned/ dried/ toasted). The first 4 columns contain the public food key, classification, derivation and food name. The other 268 columns represent the nutrients for each food that is recorded per 100g which contains energy, macronutrients, organic acids, minerals, vitamins, fatty acids, amino acids and others which we will go more into detail later.

The columns from the dataset have been summarised in Table 1 below:

Column	Description
Public Food Key	A unique alpha-numeric ID for each food
Classification	A, 2-, 3-, or 5-digit numeric classification code which groups foods hierarchically. Example, all 11xxx are non-alcoholic beverages, 111xx are teas and 11104 are herbal tea and fruit infusions
Derivation	Indicates how the majority of nutrient data, or data for key nutrients, were sourced: ● Analysed - directly measured in a laboratory ● Borrowed - sourced from another credible composition database ● Imputed / Estimated - calculated from a similar food ● Recipe - derived from a recipe calculation of constituent ingredients ● Label Data - taken from product labelling
Food Name	The name of the specific food
Nutrient name (units) (Column 5-272)	Describes the full name of each of the nutrients e.g. 'Protein' and includes the units the nutrient is presented. A value is then provided for each food and nutrient on a per 100 g basis.

Table 1: Dataset columns

2.2 Cleaning the Dataset

Prior to doing any machine learning work, it is important that we make sure that the dataset we use is free of any incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data.

Since we are only using the main dataset in this report, we first change that dataset into a csv file in order to make our processes easier.

After looking through the dataset, we found that the dataset itself is relatively clean. There are no duplicate keys, no duplicate food names, no broken Public Food Key patterns, no rogue text in numeric columns and no all-blank columns.

However, some rows have Food Name that contains a trailing space. Realistically, this “dirt” in the data will not affect much. It is still good practice to keep everything standardised.

We also discovered that most nutrient cells are blank (57%/ 243k of 425k cells to be exact). After referencing this to AFCD’s documents, it has been said that they are left blank because the nutrient was not measured for that food.

During this step, the columns that act as identifiers (Public Food Key, Food Name, Classification, Derivation) will be dropped as they do not add any new information about our target variable (Energy

with dietary fibre, equated (kJ)). Removing these columns will prevent overfitting as the models might learn unwanted patterns produced by these features.

2.3 Training and Testing Data Split

The data will then be shuffled randomly so that a 70/30 Training and Testing data split will be fair. We do this before doing further feature selections to prevent data leakage from training data to testing data. This is achieved using the train-test-split method from the scikit-learn Python package.

2.4 Handling Leaky Columns

Since we are only looking at the Energy with dietary fibre, equated (kJ) column for this report, the Energy without dietary fibre, equated (kJ) column will be nearly identical and will be labelled as a "leaky column" as it will cause leakage.

This column will be dropped from the dataset.

2.5 Handling Missing Values

As we gathered from the information above, we will drop the columns/features that are missing too much data. We will set a missingness threshold at 70% which tells us that any column that has less than 70% data will be dropped. We will only run this on our training data and the columns that have been removed will be reflected onto our testing data.

After doing this, we are left with 79 features that we will try to further reduce.

2.6 Correlation Analysis

A critical preprocessing step prior to training our model is the identification and removal of highly correlated features. In nutritional datasets such as the Australian Food Composition Database (AFCD), multicollinearity is a very common issue as there are many nutrients that are chemically or mathematically related. For instance, total saturated fatty acids is calculated by summing individual saturated fatty acid fraction. Retaining both representations of the same measurement can introduce redundancy that can affect model performance.

To address this problem, we compute the Pearson correlation coefficients between all the remaining features in the training set. We do not do it on the whole dataset or the test set as it may cause data leakage as mentioned above. Pearson's r was selected as the correlation measure because the features in this dataset are continuous and the relationships between chemically related nutrients are mostly linear. After the computation, a correlation matrix is plotted in order to show the correlations between features more intuitively. Figure 1 below shows us that correlation matrix.

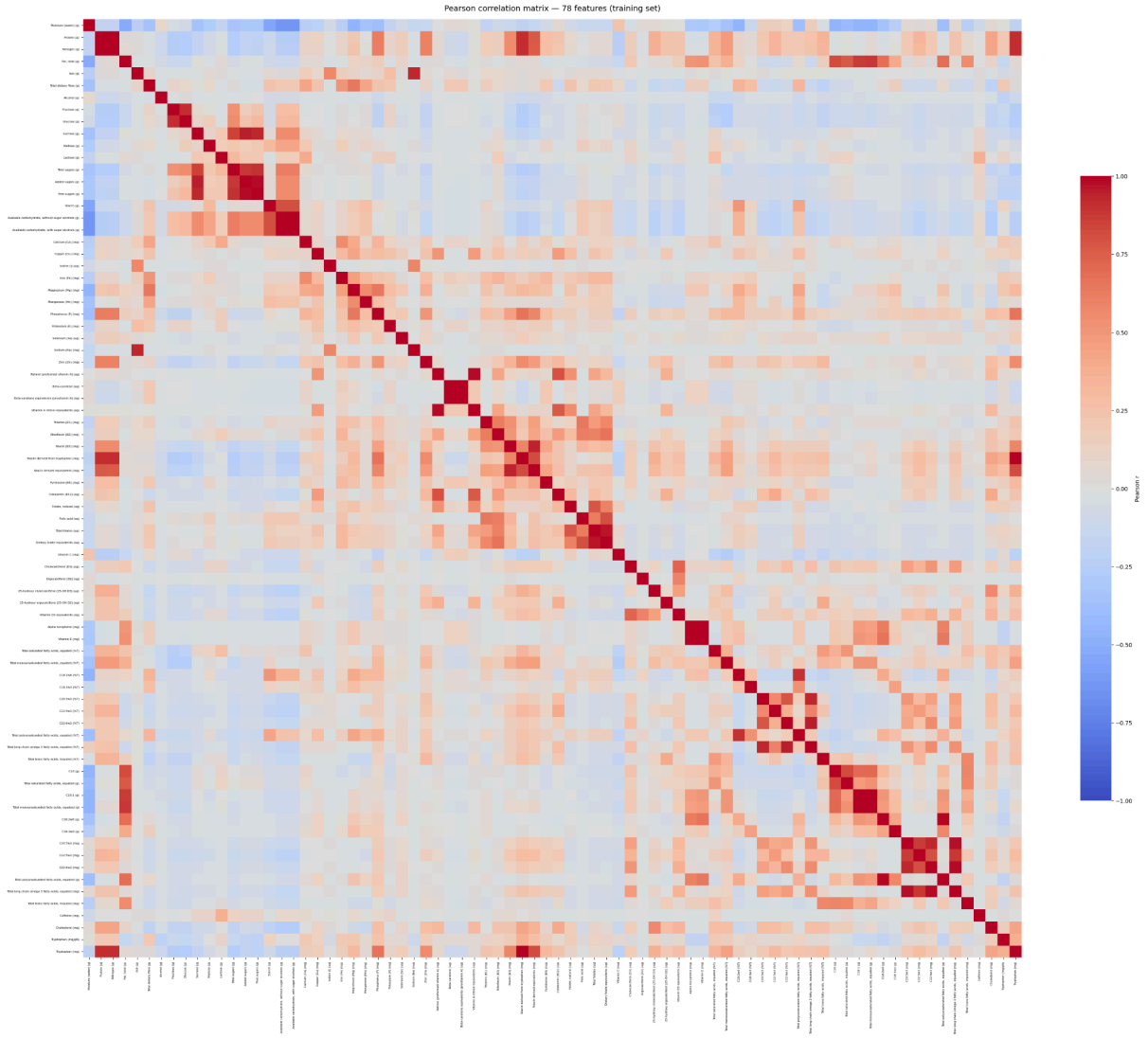


Figure 1: Correlation Matrix between Features

As we can see from the figure above, the boxes coloured deep red or deep blue tells us that the features are highly correlated (either positively correlated or negatively correlated). It is pretty difficult to look through the correlation matrix above and find the highly correlated pairs. Therefore, we have computed the correlation between features individually using the .corr() function from the python sklearn package. Table 2 below shows the top 10 feature-pairs with the highest r-values and their r-values.

Feature A	Feature B	r
Niacin derived from tryptophan (mg)	Tryptophan (mg)	1.0000
Available carbohydrate, without sugar alcohols (g)	Available carbohydrate, with sugar alcohols (g)	0.9998
Alpha tocopherol (mg)	Vitamin E (mg)	0.9962

C18:1 (g)	Total monounsaturated fatty acids, equated (g)	0.9948
Protein (g)	Nitrogen (g)	0.9928
Beta-carotene (ug)	Beta-carotene equivalents (provitamin A) (ug)	0.9923
C18:2w6 (g)	Total polyunsaturated fatty acids, equated (g)	0.9853
Retinol (preformed vitamin A) (ug)	Vitamin A retinol equivalents (ug)	0.9802
C22:6w3 (mg)	Total long chain omega 3 fatty acids, equated (mg)	0.9799
Added sugars (g)	Free sugars (g)	0.9766

Table 2: Top 10 correlated features sorted by r-value

As these feature-pairs shown above are highly correlated, we will have to drop either one of the features. This will not be done just yet as the removal of some of the features might affect our model training.

3.0 Model Training and Evaluation

3.1 Atwater System

3.1.1 Introduction

Estimating energy in the nutritional science field has always been done using linear models. The most common way to calculate energy in food is the Atwater equation. It quantifies energy in food into 4 macronutrients: protein, fat, available carbohydrates and alcohol. Each of these macronutrients are given a coefficient that is derived from caloric experiments (FAO, 2026). The equation is shown below:

$$\text{Energy (kJ)} = (\text{Protein}*17) + (\text{Fat}*37) + (\text{Available Carbohydrate}*17) + (\text{Alcohol}*29)$$

These coefficients represent the average metabolism energy yield per gram of each macronutrient while accounting for digestive losses. Most importantly, the Atwater formula requires no training data and no hyperparameter tuning and due to this each output is entirely traceable to its macronutrient inputs.

The Atwater formula will serve as a performance benchmark in this study due to the fact that it is the standard method used by the global food industry to calculate the total calories in a food product (Ball et al., 2023). Any machine learning model that fails to outperform Atwater would offer no practical advantage over a simple calculator.

3.1.2 Feature Selection for Atwater

While selecting the right features for this, we came across a problem where there are multiple columns that represent the same thing. For example, fats are represented by 'Fat, total (g)', 'Total saturated fatty acids, equated (%T)', 'Total monounsaturated fatty acids, equated (%T)' and 6 more other features. Carbohydrates have features that represent it with and without alcohol.

To simplify it, 'Protein (g)' , 'Fat, total (g)' , 'Available carbohydrate, without sugar alcohols (g)' and 'Alcohol (g)' were selected as they most closely resemble the variables needed for the Atwater formula.

As there are no missing values in either one of these features, nothing has to be done.

3.1.3 Model and Performance

Since the Atwater system does not require model training, it is applied directly to the test dataset. Figure 3 shows us the predicted vs actual energy values for all test foods. Predictions falling closer to the y = x diagonal line tells us that it was accurate.

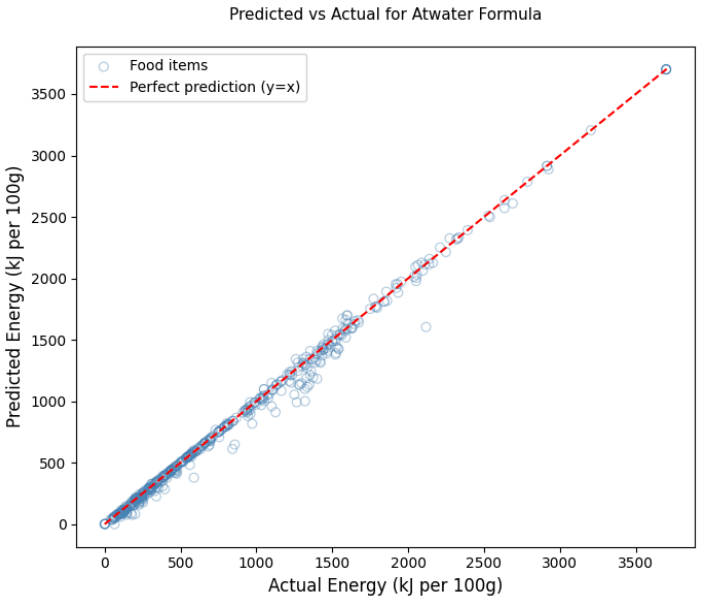


Figure 3: Predicted vs Actual for Atwater Formula

The Atwater formula achieved an R^2 of 0.9936, an RMSE of 56.67 kJ and an MAE of 28.21 kJ on the test set. This shows strong results for a rule-based system with no training phase. The gap between RMSE and MAE suggests that while Atwater performs well for most foods, it can still produce errors for a subset of food. These are likely foods with an abnormal macronutrient composition or significant fibre and organic acid content that cannot be captured by the four variables. However, this establishes a very strong benchmark for the machine learning models evaluated in subsequent sections.

To further prove our theory of the Atwater formula producing error while predicting abnormal subset of food, we plot a box plot of the absolute error of the Atwater formula based on each food category. As we know, the classification column groups the food by a 3 tier structure where it goes from 2 digit (major group level) to 3 digit (sub-major group level) to 5 digits (minor group level) (Food Standards

Australia New Zealand, 2025). For this we used the 2 digit classification level to see what major groups get the most error from the Atwater formula.

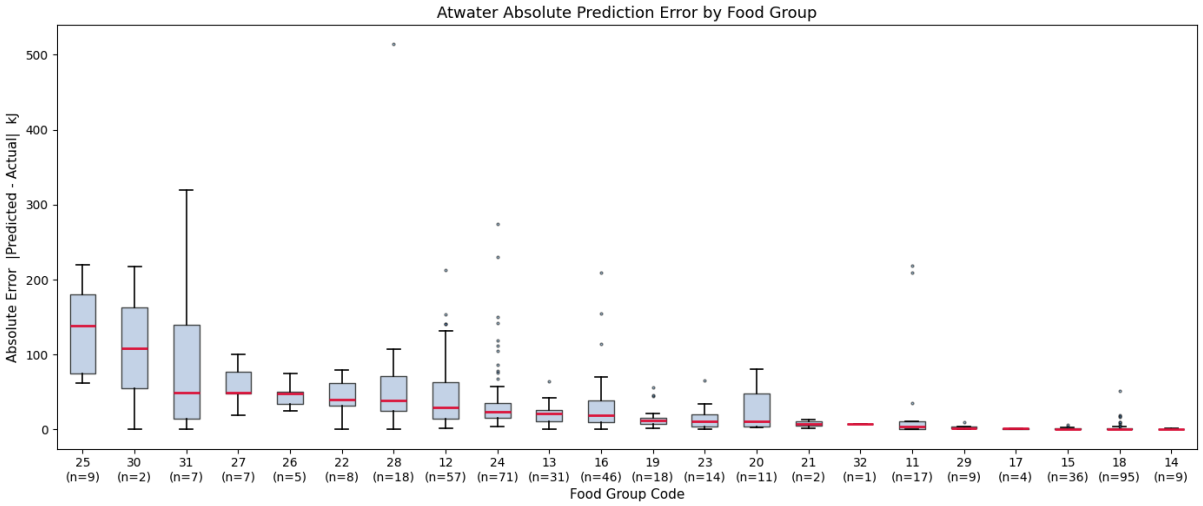


Figure 4: Atwater Absolute Prediction Error by Food Group

As we can see, the Atwater formula struggles more against groups such as 25 (legumes and pulse products), 30 (special dietary foods) and 31 (miscellaneous) as they are either more complex or might be predominantly fibre. This provides further motivation to perform machine learning to predict energy in food.

3.2 Random Forest

3.2.1 Introduction

The Atwater models evaluated in Section 3.1 assume that food energy is a straightforward weighted sum of nutrient content. This assumption holds well on average as the Atwater formula achieved an R^2 of 0.9936. However, the imbalance in Figure 4 tells us that certain food cannot be mapped linearly.

As we learnt in the week 9 lecture, Random Forest is a non-linear ensemble method that builds a large number of decision trees. These trees are each trained on a sample of the data that was bootstrapped and contains a random subset of features for each split. The predictions will be made by averaging the outputs of all the trees. This way, it reduces overfitting and produces an ensemble that can generalise better than any single tree. Unlike the linear models in Section 3.1, Random Forest can capture non-linear relationships between features without the need to specify.

Random Forest also provides us with a feature importance score that allows us to know which features are more involved in the prediction of energy. This lets us compare to the Atwater system and decide to either add onto or subtract from the formula itself. It also handles multicollinearity really well. Although we already tried preventing this problem in Section 2.6 previously by correlation analysis, a lot of features that remain are still slightly correlated due to the fact that most macro and micronutrients are related.

3.2.2 Feature Selection

As mentioned in Section 2.6. We haven't removed the correlated pairs yet since it might affect our model. However during this section, we will remove them. This leaves us with 67 features.

3.2.3 Choosing Hyperparameters

Random Forest has 2 key hyperparameters that can substantially affect model performance. These are:

- **n_estimators**: The number of trees in the forest.
- **max_features**: The number of features randomly considered at each split.

3 other parameters (max_depth, min_sample_split and min_sample_leaf) are involved in controlling the tree complexity and acts as regularization for us. We will figure out these parameters using grid search on a 5 fold cross validation using the training set only. This is to prevent any information from the test set influencing the model.

3.2.4 Model

After running cross-validation on the model, the optimum hyperparameters were n_estimators = 500, max_features = 0.33 , max_depth = 20 , and min_samples_leaf = 1. Figure 5 shows us how cross-validation RMSE changes as the number of trees increases. We can see that the performance keeps improving as more trees get added however at the 400 to 500 range, we see the performance starting to plateau. We wouldn't want to go past 500 trees as it will just increase the computation time by a lot. We are trading computation time here for a bit of performance. After deciding the hyperparameters, the full training set will be used for training the model and the held out test set will be used for calculating the evaluation metrics. The RandomForestRegressor function from the python sklearn package will be used.

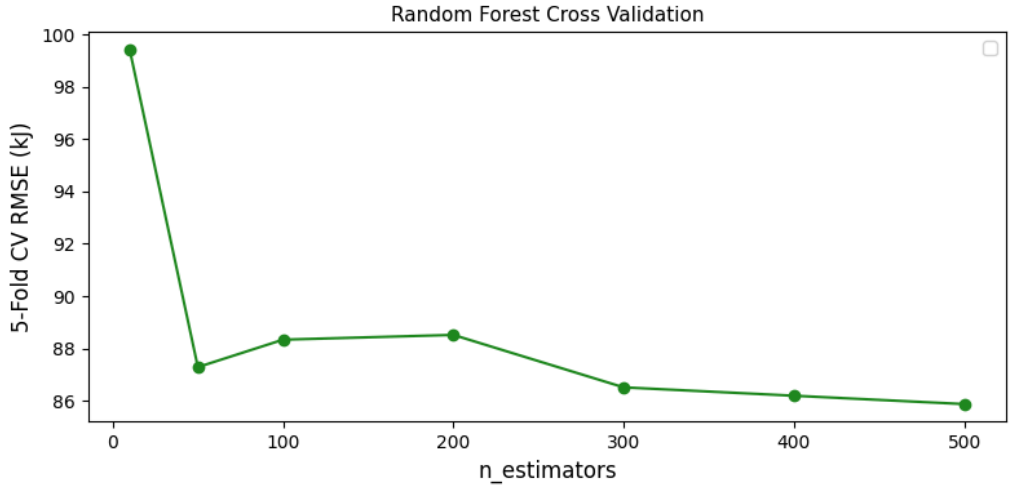


Figure 5: Random Forest Cross Validation for n_estimators

3.2.5 Performance

Random Forest achieved a R^2 of 0.9906 , an RMSE of 68.83 kJ and an MAE of 40.16 kJ on the test set. Contrary to expectations, Random Forest did not outperform the Atwater baseline. However, if we look at figure 6, the absolute error for categories that used to be really high such as 30 (special dietary food), 25 (legumes) and 31 (miscellaneous) are relatively low now. However categories such as 22 (seeds and nuts), 14 (fats and oils) and 28 (confectionery) that used to be very well predicated ended up placing higher. This tells us that random forest is better at predicting more unusual food items compared to the Atwater system.

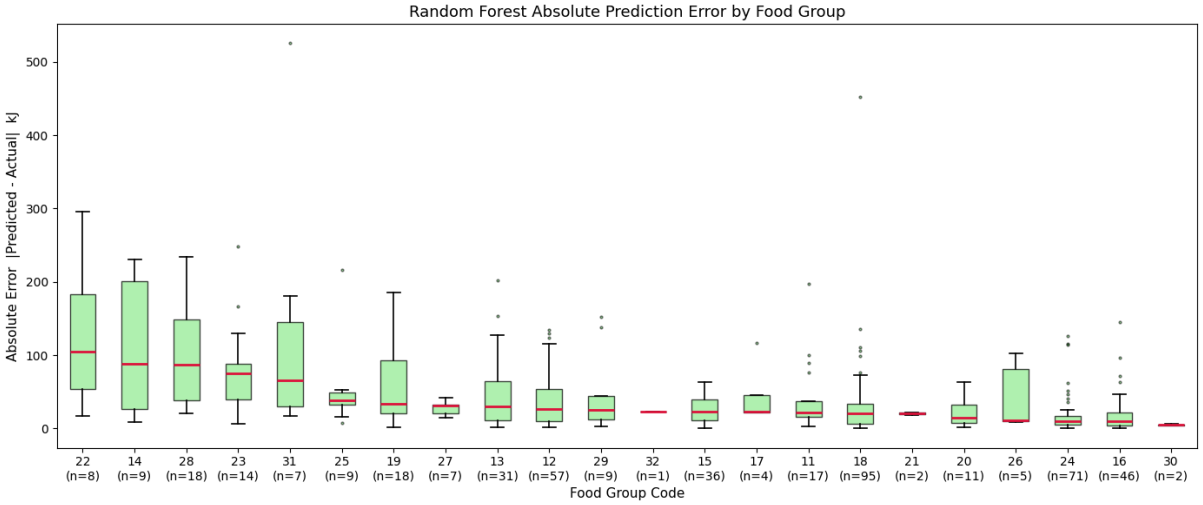


Figure 6: Random Forest Absolute Prediction Error by Food Group

Notably, the residual plot in Figure 7 down below shows us why random forest underperforms in this report. As we can see from the plot, for lower energy values, the random forest model produces errors that are closer to 0 but as we stray further away, the error starts increasing. We predict that this might be coming from the fact that the random forest model is bounded by the range of training values. Foods with very high energy are under-predicted as the model cannot predict beyond the maximum energy values seen during training. This is known as the ceiling effect. I knew from the lectures that random forest can reduce variance through bagging while keeping bias the same. However, it did not come across my mind that random forest could be bounded by the range of the training data itself and might not generalise well.

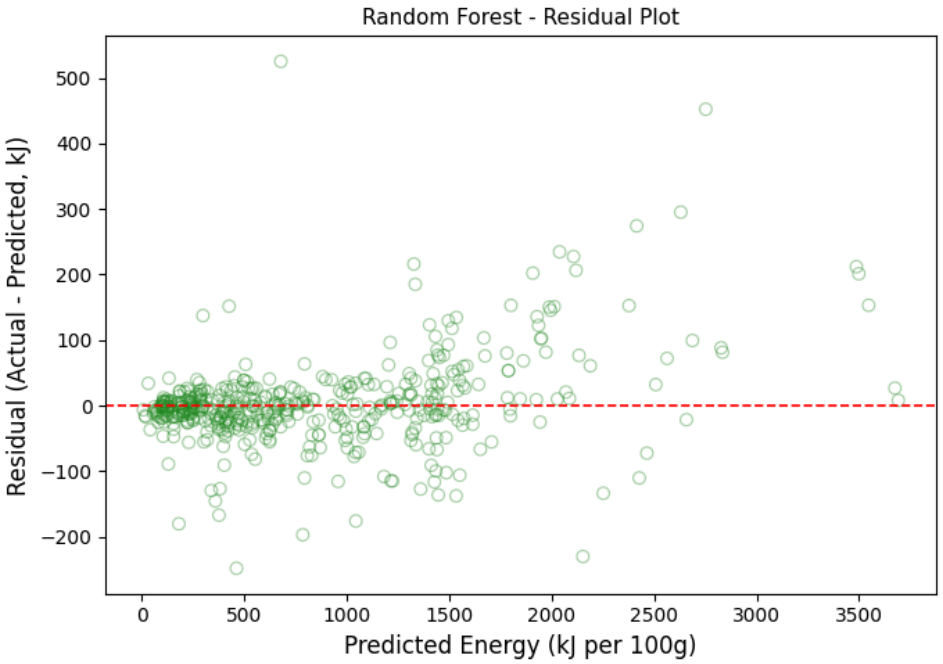


Figure 7: Random Forest residual plot

3.2.6 Feature Importance

While not providing as good of a performance as the Atwater system, random forest provides us with the ability to produce feature importance scores. As we can see from Figure 8, Moisture (g) and Fat (g) are at the top of the importance rankings. This is consistent with the fact that foods with high moisture have low energy, while dry foods have high energy (*Weight Loss: Feel Full on Fewer Calories*, 2024.). Fat is really important in predicting energy due to the fact that it is very energy dense (37 kJ/g) (Ball et al., 2023).

The most interesting fact is that carbohydrates rank quite low on feature importance and protein and alcohol are not even present in the top 15 even though it is part of the Atwater formula. This might be because most of the splits done in the random forest are taken up by moisture and fats already, not allowing carbohydrates and proteins to contribute much.

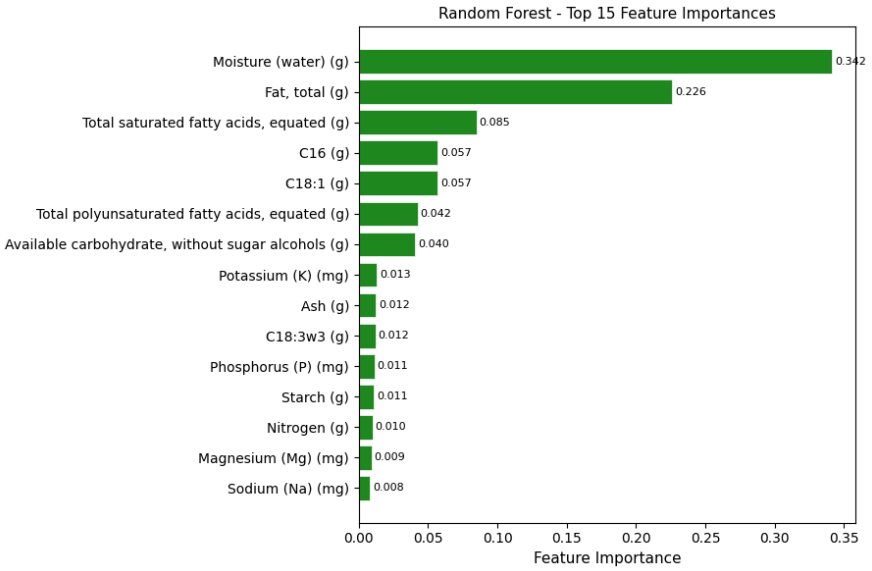


Figure 8: Top 15 ranked feature importance from random forest

3.3 XG Boost

3.3.1 Introduction

The last model we are testing here in XGBoost. It addresses one of the key limitations of Random Forest where it builds each tree independently which helps reduce variance but does not correct errors made by individual trees which leads to the ceiling effect. XGBoost uses an ensemble method called boosting. As we learnt in the lectures, this is where trees are built sequentially and each tree learns from the errors of the previous tree.

XGBoost implements gradient boosting where it uses the second order derivatives to guide tree constructions. It also has L1 and L2 regularizations to help prevent overfitting and uses columns and row subsamples to improve diversity across the trees.

3.3.2 Choosing Hyperparameters

Similar to the Random Forest model done in Section 3.2, XGBoost also requires us to have pre-defined hyperparameters before model training. However, for XGBoost there are 7 hyperparameters that we need to consider:

- **n_estimators:** The number of trees.
- **Learning_rate**: It tells us how much each tree's error contributes to the next.
- **max_depth**: The maximum depth of each tree.
- **subsample**: The fraction of training samples used for each tree.
- **colsample_bytree**: The fraction of features randomly selected for each tree.

- **reg_alpha** and **reg_lambda**: L1 and L2 regularisation penalties on leaf weights.

Similar to what we did for Random Forest, these hyperparameters are optimized by running grid search and 5 fold cross validation on the training set. However, to find the values needed for the grid search, we first run a broad random search to help us find the specific regions needed for the grid search. Early stopping was also used during cross validation, stopping when the RMSE stops improving for 20 consecutive rounds. This helps prevent overfitting.

3.3.3 Model

After optimization, the hyperparameters n_estimators = 1000 , learning_rate = 0.02 , max_depth = 4 , subsample = 0.6 , colsample_bytree = 0.8 , reg_alpha = 0.01 , and reg_lambda = 0.5 are defined. Figure 9 shows us how the learning rate was discovered. The final model was trained on the training set using the hyperparameters above and are evaluated on the testing set.

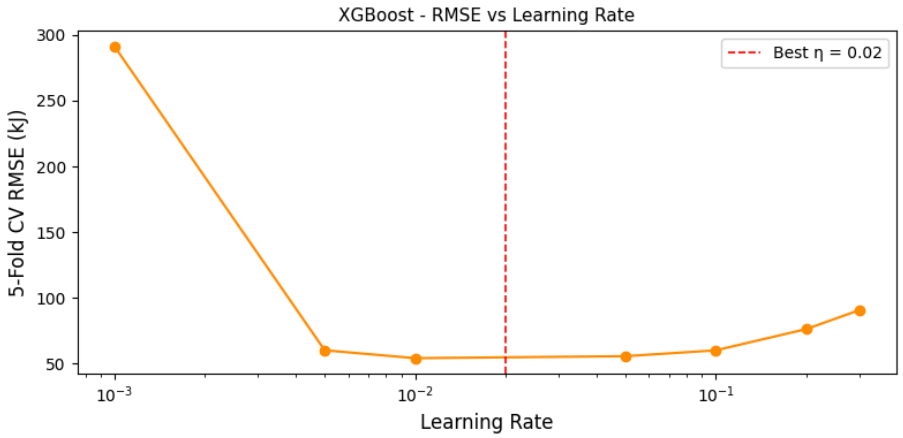


Figure 9: XGBoost Learning Rate

3.3.4 Performance

XGBoost achieved an R^2 of 0.9959, an RMSE of 45.55 kJ and an MAE of 28.16 kJ, which out-performs all the other models we tested. The improvement in RMSE was really substantial (approximately 20%) compared to the Atwater formula while the MAE was nearly identical. This tells us that both models still make mistakes on typical foods whereas XGBoost significantly reduces the large errors produced by the Atwater formula on unusual food categories. As shown in Figure 10, the predictions are all way closer to the y = x line compared to the previous models. Comparisons between the model will be done in section 4 later in the report.

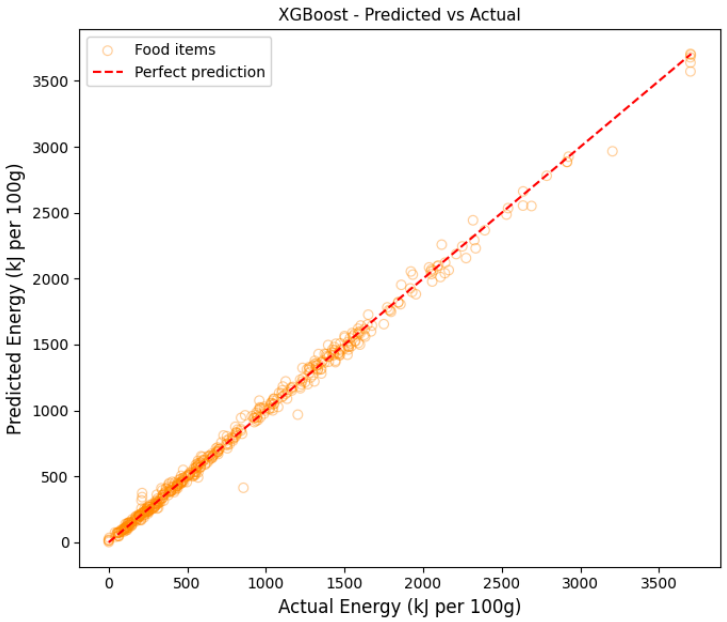


Figure 10: XGBoost, Predicted vs Actual plot

3.3.5 Feature Importance

Just like Random Forest, XGBoost also gives us a feature importance chart that is based on gain measurement. As expected, in figure 11, the XGBoost importance ranking is consistent with the rankings of Random Forest but it has fat and moisture swapped while also having carbohydrates higher up in the rankings. This helps us solidify the fact that moisture and fat are really important in measuring energy in food. It also tells us that what the Atwater formula might be missing is the calculation of moisture in food.

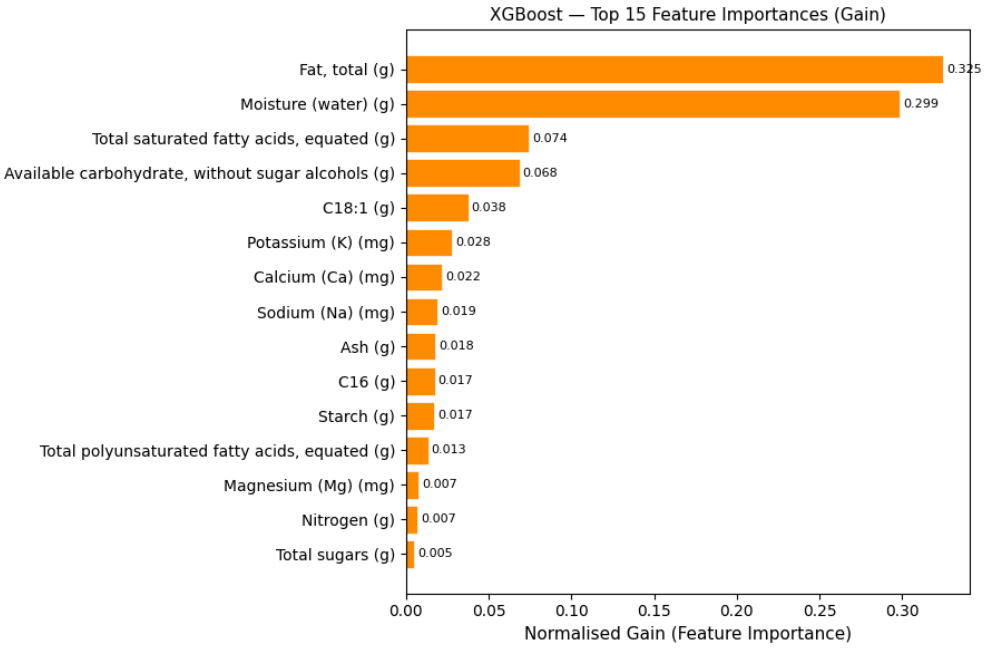


Figure 11: Top 15 feature importance XGBoost

4.0 Model Evaluation and Comparison

After training the 3 models, in this section we will evaluate and compare these models side by side. This enables us to see each model's performance more clearly and see what's wrong with each one.

4.1 Performance

Table 3 and Figure 12 summarises the performance of each model.

Model	R^2	RMSE (kJ)	MAE (kJ)
Atwater Formula	0.9936	56.67	28.21
Random Forest	0.9906	68.83	40.16
XGBoost	0.9959	45.55	28.16

Table 3: Summary of performance score for each model

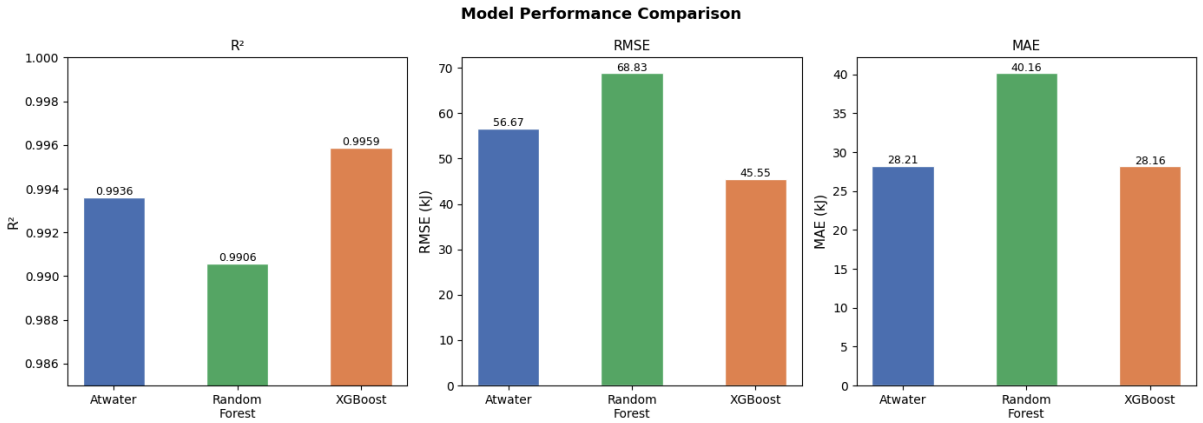


Figure 12: Bar chart showing model performance

As we can see, XGBoost achieved the strongest performance across all three metrics, with an R^2 of 0.9959, an RMSE of 45.55 kJ and an MAE of 28.16 kJ. Compared to the Atwater formula, XGBoost's RMSE is 20% less, while matching it in MAE and R^2. This nearly identical MAE tells us that XGBoost and Atwater are making similar sized errors for typical food. However, the difference in RMSE tells us that the large errors that are produced by the Atwater formula for unusual foods are not present in XGBoost. This is because RMSE penalizes large errors more heavily as shown.

4.2 Residual Analysis

Figure 13 shows us the absolute error distribution across all test foods for each model.

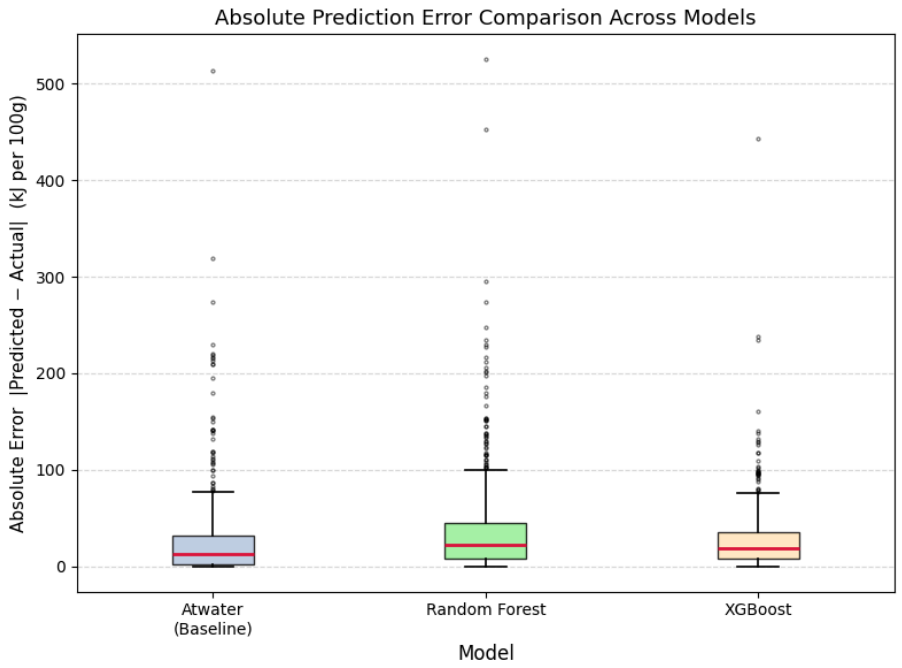


Figure 13: Absolute Prediction Error Comparison

As we can see, Atwater achieves the lowest median absolute error of 12.1 kJ while Random Forest and XGBoost got a median absolute error of 21.8 kJ and 18.5 kJ respectively. This indicates that the Atwater formula works really well with most normal foods. However, the outliers for the Atwater formula are more spread out compared to XGBoost's. This further strengthens our theory that XGBoost is more well suited for generalization than the Atwater formula.

5.0 Conclusion

The central question to this report was **Can macronutrient and micronutrient profiles alone reliably predict food energy content using machine learning?** After data cleaning, model training and model evaluations we have successfully proved that nutrient profiles alone are able to predict food energy content really well. All 3 of our models achieved a R^2 of more than 0.99 which tells us that they are all very predictive of our data.

Another thing we are trying to prove in this report is if we can outperform the century long Atwater formula. After evaluating and comparing all the models, we found that XGBoost outperforms the Atwater formula by a little bit. This is only due to the fact that XGBoost is better accommodated for abnormal food. With such a small improvement for much more computational cost, the Atwater formula still remains superior especially for routine everyday foods.

6.0 Reflection

Dealing with such large and uncleaned data helped me explore and apply different data cleaning methods learnt in the lectures and practicals. Dealing with multiple representations of the same nutrient or some nutrients that are derived from other nutrients was one of the biggest problems for

me. I've learnt in this that using correlation analysis only will not work well for that as most similar nutrients still ended up in the final feature set. Ideally, we would be able to figure out all the multiple representations and keep only one but that would require a lot.

One thing that was not expected from this report was the fact that the random forest underperformed Atwater formula. An old and simple formula with only 4 variables outperforming a machine learning model with 67 features and 500 trees was shocking to me. However, after diving deeper and finding out why this happened, it cleared up a lot in my mind. It made me realise that large and complex models don't always equal better performance.

Seeing boosting outperforming bagging connected the things learnt in lecture 9 about sequential error correction that boosting provides. Boosting techniques allow new trees to learn from the mistakes of previous trees which helped prevent the ceiling effect that was a major bottleneck for bagging. Before seeing these results, I assumed that more trees always meant better predictions regardless of the method used. Seeing XGBoost outperform Random Forest specifically on the high-energy foods made the difference between these 2 ensemble methods feel more real rather than just a definition on a paper.

This report also made me realize how different all the evaluation metrics are. I always thought that RMSE and MAE are both interchangeable since it only measures error. However, using both in this report showed me that they both are responsible for different things. MAE captured how each model performed on typical foods, while RMSE exposed how badly Atwater failed on unusual categories. I definitely would not have figured this difference out just by reading the notes itself, doing this report had made me realize that.

7.0 References

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8.0 Appendix

8.1 Features left after correlation analysis

Features
Moisture (water) (g)
Nitrogen (g)
Fat, total (g)
Ash (g)
Total dietary fibre (g)
Alcohol (g)
Fructose (g)
Glucose (g)
Sucrose (g)
Maltose (g)
Lactose (g)
Total sugars (g)
Added sugars (g)
Starch (g)
Available carbohydrate, without sugar alcohols (g)
Calcium (Ca) (mg)
Copper (Cu) (mg)

Iodine (I) (ug)
Iron (Fe) (mg)
Magnesium (Mg) (mg)
Manganese (Mn) (mg)
Phosphorus (P) (mg)
Potassium (K) (mg)
Selenium (Se) (ug)
Sodium (Na) (mg)
Zinc (Zn) (mg)
Retinol (preformed vitamin A) (ug)
Beta-carotene equivalents (provitamin A) (ug)
Thiamin (B1) (mg)
Riboflavin (B2) (mg)
Niacin (B3) (mg)
Niacin derived equivalents (mg)
Pyridoxine (B6) (mg)
Cobalamin (B12) (ug)
Folate, natural (ug)
Folic acid (ug)
Dietary folate equivalents (ug)
Vitamin C (mg)
Cholecalciferol (D3) (ug)
Ergocalciferol (D2) (ug)
25-hydroxy cholecalciferol (25-OH D3) (ug)
25-hydroxy ergocalciferol (25-OH D2) (ug)
Vitamin D3 equivalents (ug)
Vitamin E (mg)

Total saturated fatty acids, equated (%T)
Total monounsaturated fatty acids, equated (%T)
C18:2w6 (%T)
C18:3w3 (%T)
C20:5w3 (%T)
C22:5w3 (%T)
C22:6w3 (%T)
Total polyunsaturated fatty acids, equated (%T)
Total long chain omega 3 fatty acids, equated (%T)
Total trans fatty acids, imputed (%T)
C16 (g)
Total saturated fatty acids, equated (g)
C18:1 (g)
C18:3w3 (g)
C20:5w3 (mg)
C22:5w3 (mg)
C22:6w3 (mg)
Total polyunsaturated fatty acids, equated (g)
Total trans fatty acids, imputed (mg)
Caffeine (mg)
Cholesterol (mg)
Tryptophan (mg/gN)
Tryptophan (mg)